

COURSE DESCRIPTION			
Program	Biomedical Engineering, Electronics and Communication, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Integrated Circuit Design & Fabrication, Instrumentation Engineering, Micro Electronics		
Course Title	Elementary Concepts of Electronics		
Course Code	1041	Semester	I
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	100

Introduction

This course provides foundational knowledge of electronics required for diploma engineering programs. It covers basic electrical quantities, electronic signals, and the use of measuring instruments essential for understanding electronic systems. The course introduces passive and active electronic components, along with semiconductor theory, to enable analysis of basic electronic circuits. It also emphasizes electrical safety, printed circuit board concepts, and soldering fundamentals necessary for safe and effective handling of electronic systems. The course prepares learners for advanced electronics subjects and practical applications.

Course Objectives

1	To introduce fundamental concepts of electronics, electrical quantities, signal types, and electronic measuring instruments.
2	To develop understanding of passive electronic components and their configurations in basic electronic circuits.
3	To explain the principles of semiconductor devices and their applications in electronic systems.
4	To introduce PCB technology, basic PCB fabrication methods, and soldering techniques used in electronic circuit assembly.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic Knowledge in Mathematics			Secondary Education
Basic knowledge in Physics			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Demonstrate basic concepts of electronics, electrical quantities, signal types, and electronic measuring instruments in simple electronic systems.	Apply
CO2	Construct and analyze basic electronic circuits using passive electronic components and their configurations.	Apply
CO3	Apply semiconductor theory to demonstrate the operation, characteristics and applications of basic semiconductor devices.	Apply
CO4	Demonstrate Printed Circuit Boards and apply soldering techniques.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	3	2	1	-	-	-	-	1
CO2	3	2	2	3	2	1	-	-	-	-	1
CO3	3	2	2	3	2	-	-	-	-	-	1
CO4	3	2	1	2	2	-	-	-	-	-	1
Total	12	5	6	11	8	2	-	-	-	-	4
Correlation Level	3	1.25	1.5	2.67	2	1	-	-	-	-	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
3	0	3	0	6	90	4.5	20	60	80	20	40	60
<i>L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination</i>												

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Introduction to Electronics, Signals and Measurements	Introduction to Electronics - Applications - Basics of Electrical Quantities - Signals - Electronic Measuring Instruments	11	0
Module I	Introduction to Electronics, Signals and Measurements	Familiarization with Electronics labs - Identification of Electronic systems used in daily life Identification of Electrical Sources - Identification of basic electronic instruments - Measurement of parameters of sinusoidal ac signals	0	6
Module II	Passive Electronic Components	Resistors - Series and parallel configurations - Capacitors - Series and parallel configuration - Inductors - Transformers	12	0
Module II	Passive Electronic Components	Identification and testing of passive components - Value verification of resistance and capacitance - Series and parallel resistor circuits - Series and parallel capacitor circuits - Voltage divider circuit - RC timing circuits	0	9
Module III	Active Electronic Components	Energy Band Diagrams - Intrinsic semiconductor - Doping and Extrinsic semiconductors - Formation of a PN Junction Diode - Forward and reverse bias - VI Characteristics and Diode ratings - Zener Diode -LED - Photo diode	12	0
Module III	Active Electronic Components	Identification of PN Junction Diode and study of biasing - Plotting V-I characteristics of PN Junction Diode - Plotting VI Characteristics of Zener Diode	0	9
Module IV	PCB and soldering	Printed Circuit Board (PCB) - Types of PCBs - PCB Materials - Basic PCB Fabrication Methods - Soldering	10	0
Module IV	PCB and soldering	Identification of PCB Types and PCB Materials - Soldering Practices.	0	12

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to Electronics	Define electronics and identify its basic purpose in engineering applications.	L	CO1	Understand	0.5
Introduction to Electronics	Outline the major developments in the evolution of electronics.	L	CO1	Understand	0.5
Familiarization of Electronics labs - Identification of Electronic systems used in daily life	Familiarize with electronics laboratories and recognize common electronic systems and equipment used in daily life.	P	CO1	Understand	3
Applications	Describe the importance of electronics in modern engineering fields. Discuss importance of electronics in communication systems, industrial automation and control systems, medical and healthcare systems, instrumentation and measurement systems, automobiles, consumer electronics, defense and security systems, computer and IT systems, and renewable energy systems.	L	CO1	Understand	2
Basics of Electrical Quantities	Explain the concepts of electric charge, electric current, electric voltage, electric power, and electrical energy along with their units.	L	CO1	Understand	2
Basics of Electrical Quantities	Differentiate between AC and DC quantities in terms of their characteristics. List practical aspects of AC and DC quantities .	L	CO1	Understand	1

Signals	Explain the concept of a signal. Describe analog and digital signals.	L	CO1	Understand	1
Signals	Explain the AC waveform and its equation. Describe fundamental parameters of alternating quantities- cycle, time period, frequency, angular velocity, amplitude (instantaneous value, and peak value).	L	CO1	Understand	2
Electronic Measuring Instruments	Discuss the need for measuring instruments in electronics. Explain the applications of voltmeter, ammeter, multimeter, DC power supply, function generator, and digital storage oscilloscope (DSO).	L	CO1	Understand	2
Identification of Electrical Sources	Identify and classify electrical sources as AC or DC. Operate a line tester to identify phase and neutral lines of a single phase supply.	P	CO1	Apply	1
Identification of basic electronic instruments	Demonstrate basic electronic measuring instruments.	P	CO1	Apply	1
Measurement of parameters of sinusoidal ac signal	Measure parameters of a sinusoidal AC signal.	P	CO1	Apply	1
Module II					
Resistors	Describe the symbol, working principle, parameters, and applications of resistors. Compare the features of fixed resistors - carbon composition, carbon film, metal film, and wire-wound resistors.	L	CO2	Understand	1
Resistors	Identify and explain the operation and applications of variable resistors - presets and potentiometers. Discuss working principle of special resistors - LDR, VDR, and thermistor.	L	CO2	Understand	1
Resistors	Determine the resistance value and tolerances using resistor colour coding	L	CO2	Apply	1
Series and parallel resistor configuration	Explain the concepts of series and parallel resistor configurations. Calculate equivalent resistance.	L	CO2	Apply	2
Capacitors	Describe the symbol, working principle, parameters, and charging behavior of capacitors.	L	CO2	Understand	1
Capacitors	Describe the types of capacitors - fixed polarized (electrolytic), fixed non-polarized (ceramic, film/paper, mica), and variable capacitors (ganged capacitor, padder, trimmer). List their applications.	P	CO2	Understand	1
Identification of Passive Components	Identify passive electronic components.	P	CO2	Understand	1
Verification of resistance and capacitance Value	Determine resistor values using colour coding and verify the measured resistance using a multimeter.	P	CO2	Apply	1
Verification of resistance and capacitance Value	Determine capacitance values using direct marking, numeric coding, and colour coding, and verify them using a multimeter.	P	CO2	Apply	1
Series and parallel capacitor configurations	Demonstrate the concepts of series and parallel capacitor configurations to find equivalent capacitance.	L	CO2	Apply	1
Series and Parallel Resistor Circuits	Construct series and parallel resistor circuits and measure the equivalent resistance using measuring instruments.	P	CO2	Apply	1.5
Series and Parallel Capacitor Circuits	Construct series and parallel capacitor circuits and measure the equivalent capacitance using measuring instruments.	P	CO2	Apply	1.5
Inductors	Describe the symbol, construction (air core inductor), working principle, parameters. List applications	L	CO2	Understand	1
Transformers	Describe the construction and working principle of a transformer.	L	CO2	Understand	2

Transformers	Describe the types of transformers - step-up, step-down, and isolation transformers. List applications.	L	CO2	Understand	1
Voltage divider circuit	Construct voltage divider circuit and verify its operation through measurements and observation.	P	CO2	Apply	1.5
RC timing circuits	Construct RC timing circuit and verify its operation through measurements and observation.	P	CO2	Apply	1.5
	Lab Series Exam - I		CO2	Apply	3
Module III					
Energy band diagrams	Describe the energy band diagram of conductor, insulator, and semiconductor.	L	CO3	Understand	0.5
Intrinsic semiconductor	Explain the concept of intrinsic semiconductors and thermal generation of charge carriers.	L	CO3	Understand	0.5
Doping & extrinsic semiconductors	Explain the concept of doping in semiconductors and the formation of N-type and P-type semiconductor	L	CO3	Understand	1
Doping & extrinsic semiconductors	Explain the role of majority and minority charge carriers in extrinsic semiconductors.	L	CO3	Understand	1
Formation of a PN junction	Explain the formation of a PN junction, the development of the depletion region, and the roles of drift and diffusion currents. Describe diode with symbol.	L	CO3	Understand	2
Identification of a PN Junction Diode	Apply diode testing methods to identify terminals of a PN junction diode and classify diodes based on material.	P	CO3	Apply	3
Forward and reverse biasing of PN junction diode	Understand the effect of forward and reverse biasing on a PN junction diode	L	CO3	Understand	2
V-I characteristics of a PN junction diode	Explain the V-I characteristics, operating parameters, and ratings of a PN junction diode.	L	CO3	Understand	1
V-I characteristics of a PN junction diode	Plot the V-I characteristics of a PN junction diode and determine its parameters - static and dynamic resistance.	P	CO3	Apply	3
Zener diode	Explain the Zener diode, Symbol, V-I characteristics. List applications.	L	CO3	Understand	2
V-I Characteristics of a zener diode	Plot V- I characteristics of zener diode	P	CO3	Apply	3
LED	Describe the working principle of Light Emitting Diodes, symbol. List applications	L	CO3	Understand	1
Photodiode	Describe the working principle of a photodiode, symbol. List applications.	L	CO3	Understand	1
Module IV					
Printed Circuit Board (PCB)	Explain the purpose and basic structure of printed circuit boards (PCBs).	L	CO4	Understand	1
PCB Types	Classify different types of PCBs - single-sided, double-sided, and multilayer PCBs. List applications.	L	CO4	Understand	1
PCB Materials	List common materials used for PCB fabrication.	L	CO4	Remember	1
PCB Layout and Design Basics	Understand PCB layout concepts including track, pad, component placement, and spacing requirements.	L	CO4	Understand	2
Basic PCB Fabrication Method	Explain the basic steps involved in simple PCB fabrication including layout preparation, etching, drilling, and component mounting	L	CO4	Understand	3
Identification of PCB Types and PCB Materials	Identify PCB types and PCB materials used in electronic circuits	P	CO4	Apply	3
Soldering	Explain basic soldering concepts, materials and the use of soldering tools.	L	CO4	Understand	2
Soldering Practices	Apply soldering techniques to make reliable joints	P	CO4	Apply	9
	Lab Series Exam - II	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Identify domestic electronic systems and list their purpose.	6
2	Generate a sine wave using a Function Generator. Measure frequency using Function generator display and DSO measurement / cursor method. Record both readings and calculate percentage of error if any.	6
3	Collect different types of batteries (minimum 3 types), measure their voltages using a suitable instrument, and compare the measured values with the rated voltages.	6
4	Download or collect user manuals/datasheets of any one instrument (Multimeter / DSO / Function Generator / DC Power Supply)	6
Module II		
5	Download datasheets of any one resistor and one capacitor. Identify Rating, Tolerance, Power / voltage rating, Package type	6
6	Take 3 capacitor values. Calculate equivalent capacitance for Series connection, and Parallel connection. Verify the value with multimeter	6
7	Observe any step-down transformer used in daily life such as a mobile charger, power adapter, or electronic power supply. (No dismantling is required.) From the nameplate or printed specifications, record primary rating (Input voltage and frequency), Secondary rating (Output voltage and current)	6
8	Visit an electric vehicle charging station. Prepare a document about the specifications of chargers available there.	6
Module III		
9	Prepare a comparative study of different types of light sources (Incandescent lamp, Fluorescent tube, LED, Halogen, etc)	6
10	Prepare a chart on evolution of semiconductor devices	6
11	Explore different types of circuit simulation tools available online. Write a document that explore features of any one	6
Module IV		
12	Prepare a document that describes the technical features of a mobile phone (Hardware and software features)	6
13	Prepare a chart or poster that describes different types of soldering iron.	6
14	Observe the PCB inside any electronic product (adapter/ remote/ charger - without dismantling if sealed). Identify: (1) PCB type, (2) Approximate size, (3) Mounted components (resistor, diode, IC, capacitor)	6
15	Study simulation software like tinker cad. Simulate a simple circuit to turn on and off an LED with a switch. (https://www.tinkercad.com)	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Electronic Inventions and Discoveries: Electronics from its earliest beginnings to the present day, Fourth Edition	G.W.A Drummer	Pergamon
2	Grob's Basic Electronics	Mitchel E. Schultz	McGraw-Hill
3	Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S. I. Units	B.L. Theraja and A. K. Thereja	S. Chand
4	Electronic devices and circuit theory	Robert Boylestad and Louis Nashelsky	PHI
5	Practical electronics for inventors	Paul Scherz and Simon Monk	McGraw-Hill
6	A Text Book of Applied Electronics	R S Sedha	S Chand
7	Basic Electronics and Linear Circuits	N N Bhargava, Kulshreshtha and S C Gupta	McGraw-Hill

REFERENCE			
Sl. No.	Title	Author	Publication
1	Electronic Devices and Circuits	Robert Boylestad	PHI

2	Electronic Devices and Circuits	David A Bell	PHI
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WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/122106025	I, II & III
2	https://onlinecourses.swayam2.ac.in/ntr25_ed130/preview	III
3	http://vlabs.iitkgp.ac.in/be/#	II & III
4	https://onlinecourses.swayam2.ac.in/nou26_ee01/preview	IV
5	https://onlinecourses.nptel.ac.in/noc24_ee127/preview	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Digital Storage Oscilloscope	3 MHz, Dual channel	15
2	Signal/ Function Generator	3 MHz minimum	15
3	Power Supply Unit	30V, Dual	15
4	Multimeter	DMM (with capacitor measurement mode)	15

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		15	50	40	40	50	50	60	40
Converted to		5	10	10	10	10	10		
Marks	5	5	10	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60